Priced to Stand Still: The Missing Cutting Cycle in Brazil's Local Yield Curve

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ABSTRACT

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I fit Nelson–Siegel–Svensson curves to free public data for four sovereign local-currency markets (Brazil, Mexico; United States and Germany as anchors) and read each curve's short end as a forward-implied policy path. Across the set, Brazil alone prices no easing: the implied 3-month rate nine months forward sits −8.4 bp from a 14.25% Selic, while every other curve is steep to its own five-year history. I express the dislocation as a DV01-neutral 5s10s steepener carrying +2.8 bp per quarter per unit of DV01, and quantify its risks — a retail-quote basis, a fat flattening tail, and the term-premium content of the path metric. The analytics, data, and this paper reproduce from a public repository. [compress to 150 words or fewer in your voice]

Keywords: term structure, Nelson–Siegel–Svensson, emerging markets, local-currency sovereign bonds, monetary policy expectations, curve trades.

JEL classification: E43, E52, G12, G15, C13.

1 Introduction

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Motivate the question: local-currency EM curves encode a policy path, and cross-market dispersion in that path is tradable. State the finding — Brazil prices −8.4 bp of change over 12m against a 14.25% Selic while its 5s10s is flat (z −0.56) — and the contribution: a reproducible, free-data pipeline that ranks curve trades by carry per unit of vol. Cite the slope-factor tradition (Litterman and Scheinkman 1991) and the fitted-curve lineage (Nelson and Siegel 1987), (Svensson 1994), and (Diebold and Li 2006). [your framing and positioning]

2 Data

I fit government curves in the tradition of (Gürkaynak et al. 2007), using only free, public sources; each fetcher fails loudly rather than returning partial data, and all series are cached so results reproduce offline. Brazilian prefixado yields come from the Tesouro Direto daily rates file (LTN and NTN-F), a retail window: on the one cross-check date the public ANBIMA interbank curve allows, the fitted retail curve sits −7.6 bp, −23.1 bp, and −19.3 bp below interbank at 1, 5, and 10 years. This is a level effect that cancels in z-scores; the tenor differential is carried as a risk (Section 4). Mexican yields are Banxico weekly auction results, so no single date holds a full curve; each tenor is forward-filled at most 45 business days, and the resulting staleness is tested in Section 5. US par yields are from the Treasury and German Svensson parameters from the Bundesbank. Policy rates are normalized to the same effective-annual basis as the curves. Appendix A details every source and convention; Appendix B reports fit quality.

3 Methodology

For each date I fit the Nelson–Siegel–Svensson (NSS) forward-rate curve. The zero yield at maturity τ is

$$y(\tau) = \beta_0 + \beta_1 \frac{1 - e^{-\tau/\lambda_1}}{\tau/\lambda_1} + \beta_2\left(\frac{1 - e^{-\tau/\lambda_1}}{\tau/\lambda_1} - e^{-\tau/\lambda_1}\right) + \beta_3\left(\frac{1 - e^{-\tau/\lambda_2}}{\tau/\lambda_2} - e^{-\tau/\lambda_2}\right), \tag{1}$$

reducing to Nelson–Siegel ($\beta_3 = 0$) where fewer than six tenors are observed. Conditional on the decay parameters λ_1, λ_2, equation (Equation 1) is linear in β, so I grid the decays over a bounded range and solve ordinary least squares at each node, selecting the fit that minimizes root-mean-square error. The forward-implied policy path reads the short end as expected policy. With $f(t_1, t_2)$ the implied forward between horizons,

$$f(t_1, t_2) = \left(\frac{(1 + y(t_2))^{t_2}}{(1 + y(t_1))^{t_1}}\right)^{\frac{1}{t_2 - t_1}} - 1, \tag{2}$$

I measure easing priced over twelve months as $f(0.75, 1) - r$ for policy rate r, replacing the cruder $2(y(1) - r)$ linear-path proxy (compared in Section 4). A DV01-neutral slope trade sizes the short-tenor leg by

$$\frac{N_s}{N_l} = \frac{\mathrm{DV01}_l}{\mathrm{DV01}_s}, \quad \mathrm{DV01}(y, T) = \frac{TP}{1+y} \times 10^{-4}, \tag{3}$$

with P the zero price. Static 3-month carry-plus-rolldown on a leg of tenor T, funded at policy rate r, in bp per unit DV01, is

$$c(T) = [h(y(T) - r) + (T - h)(y(T) - y(T - h))] \times \frac{10^4}{T}, \quad h = 0.25, \tag{4}$$

and the trade value is the DV01-neutral difference of its legs. I scale each trade by the annualized standard deviation of daily fitted-slope changes and rank on carry per unit of vol. The slope z-score compares the current fitted slope to its trailing five-year mean in units of that window's standard deviation.

4 Results

Table 1 reports the cross-market state. Brazil is the only market pricing easing over the next year (−8.4 bp) and the only 5s10s below its five-year mean (z −0.56); Mexico's 2s10s offers the highest carry per unit of vol (0.24) but enters +1.31 standard deviations steep. Figure 1 places each trade in policy-path/z-score space; Figure 2 ranks carry per unit of vol; Figure 3 shows the fitted curves against six months prior.

Trade	Policy %	Path 12m bp	Slope bp	z (5y)	C+R bp/3m	Vol bp	C/V	Exit bp
MX 2s10s	6.81	+105	+139	+1.31	+17.2	71	0.24	+26
MX 5s10s	6.81	+105	+50	+1.00	+6.4	47	0.14	+11
US 2s10s	3.88	+49	+43	+0.52	+3.4	26	0.13	−13
US 5s10s	3.88	+49	+31	+0.44	+0.2	18	0.01	+10
BR 2s10s	14.25	−8	+50	+0.29	+4.1	61	0.07	−6
BR 5s10s	14.25	−8	+4	−0.56	+2.8	50	0.06	−17
DE 2s10s	2.31	+56	+44	+0.42	+3.4	18	0.19	−4
DE 5s10s	2.31	+56	+34	+0.52	+0.1	10	0.01	+17

Table 1. Cross-market state. C+R is 3-month carry plus rolldown for the steepener, per unit DV01; Exit is the one-sigma invalidation level for the positive-carry direction. Policy rates effective annual.

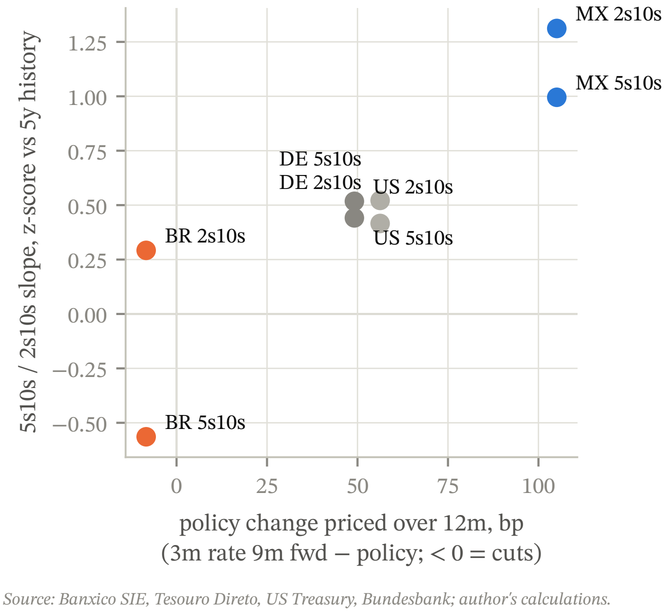
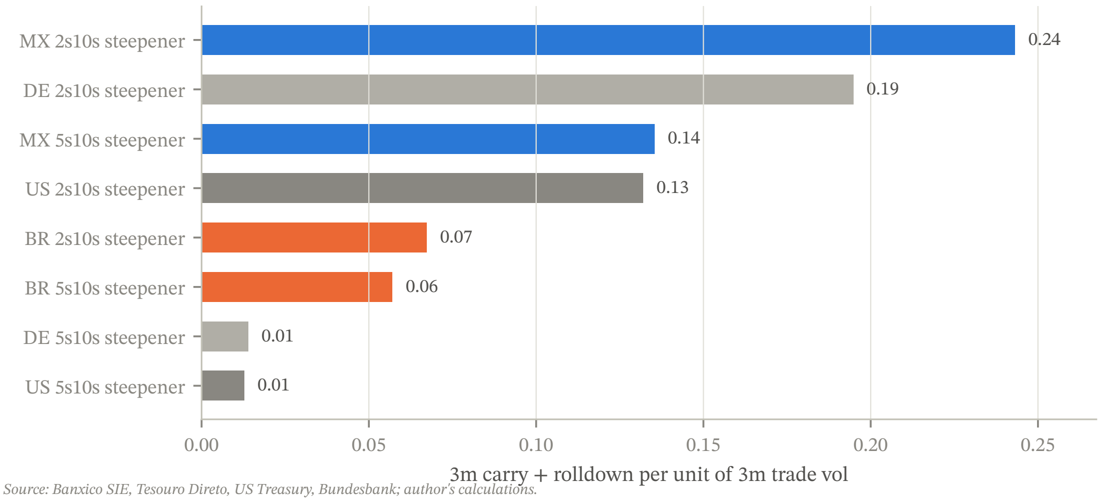
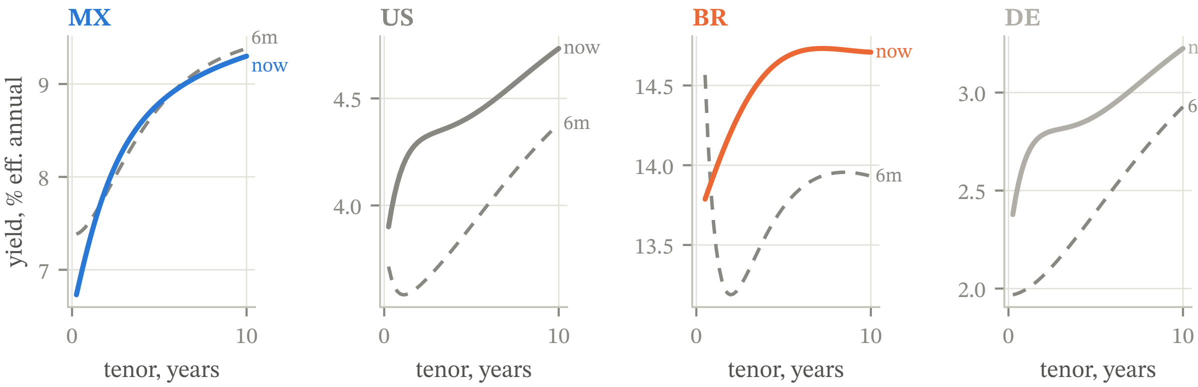


Figure 1. Policy change priced over twelve months (forward-implied) against the 5s10s / 2s10s slope z-score. Brazil alone occupies the no-cuts, flat-curve quadrant.



Source: Banxico SIE, Tesouro Direto, US Treasury, Bundesbank; author's calculations.

Figure 2. Carry plus rolldown per unit of 3-month trade vol, each trade shown in its positive-carry direction.



Source: Banxico SIE, Tesouro Direto, US Treasury, Bundesbank; author's calculations.

Figure 3. Fitted local curves today versus six months prior. Brazil un-inverted at the front, moving the implied cutting cycle outward.

The naive and forward-implied path metrics diverge most for Brazil (−62.1 bp versus −8.4 bp); the forward metric is the curve's statement about the policy rate a year out, and the cross-market ranking is robust to the choice (Appendix C).

5 Robustness

I run four checks; none constructs a profit-and-loss backtest.

Decay sensitivity. I refit each curve with the Nelson–Siegel decay pinned at the 10th, 50th, and 90th percentiles of its own fitted-decay history and recompute the trade quantities (Table 2). The Brazil 5s10s carry is nearly invariant, ranging 1.6–1.8 bp across the full decay range; the slope and z-score move more, since pinning the decay away from its fitted value re-shapes the belly. The Mexico 2s10s is more decay-sensitive, as its front leg sits where the decay term is largest.

Country	Trade	Pctile	λ (yrs)	Slope bp	z (5y)	Carry bp
BR	5s10s	p10	0.40	+13.8	+0.47	+1.6
BR	5s10s	p50	0.76	+16.3	+0.07	+1.7
BR	5s10s	p90	1.86	+13.2	−0.38	+1.8
MX	2s10s	p10	0.34	+98.6	+1.42	+26.0
MX	2s10s	p50	1.55	+147.7	+1.36	+16.4
MX	2s10s	p90	5.00	+171.9	+1.43	+13.4
US	5s10s	p10	0.37	+9.8	+0.72	+2.5
US	5s10s	p50	2.03	+32.3	+0.55	+2.0
US	5s10s	p90	4.18	+31.7	+0.84	+1.5

Table 2. Latest slope, z-score, and carry with the decay pinned at percentiles of each curve's fitted-decay history.

Fit stability. The 66-business-day rolling median RMSE (Table 3) stays near its full-sample median outside stress. Brazil's worst window reaches 14.3 bp in Feb 2021, roughly four times its median, so fitted-slope readings from that period carry wider error.

Curve	Median RMSE bp	Worst 66d RMSE bp	Worst window
MX	9.0	16.2	Oct 2022
US	3.5	8.4	Oct 2022
BR	3.4	14.3	Feb 2021

Table 3. Full-sample median versus worst 66-day rolling-median fit RMSE, by curve.

Sampling frequency. Weekly- and daily-sampled slope vols agree for Mexico (ratio near one), confirming that the 45-day forward-fill does not damp the daily vol estimate; for the daily sources the weekly figure is lower, so the daily estimate is, if anything, conservative in the sizing denominator (Table 4).

Country	Trade	Vol daily bp	Vol weekly bp	Weekly / daily
MX	2s10s	70.8	71.1	1.00
MX	5s10s	46.9	47.1	1.00
US	2s10s	25.7	18.1	0.71
US	5s10s	18.3	12.4	0.68
BR	2s10s	60.8	50.8	0.84
BR	5s10s	49.6	34.9	0.70
DE	2s10s	17.6	17.4	0.99
DE	5s10s	10.2	9.4	0.92

Table 4. Daily- versus weekly-sampled 3-month slope vol, per trade.

Mean reversion. Pooling all eight trades, I count every date whose trailing five-year z-score exceeds one in absolute value and ask whether the slope moves toward its mean over the next 66 business days (Table 5). Reversion is asymmetric: from steep levels ($z > +1$) the slope flattens 0.63 of the time across 1450 events, but from flat levels ($z < -1$) it steepens only 0.49 of the time across 3258 events, near a coin flip. The Brazil trade is a flat-slope position, so its edge rests on the forward-implied path mispricing and positive carry, not on z-score mean reversion; the z-score is context, and the invalidation level is risk management rather than expected alpha.

Bucket	Events	Toward-mean hit rate	Median move bp
$z > +1$ (steep)	1450	0.63	+5.5
$z < -1$ (flat)	3258	0.49	−0.7

Table 5. Count-based mean-reversion check, pooled across all trades. Hit rate is the fraction of dates whose slope moves toward its mean over the next 66 days; 50% is the no-reversion benchmark.

6 Discussion and limitations

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Interpret the dislocation and its likely resolution; discuss the retail basis (−23.1 bp at 5y), the term-premium content of the path metric, the par-versus-zero approximation, and the policy-rate funding assumption. State what desk data (DI futures, interbank ETTJ history, option-implied vol) would sharpen. [your interpretation and caveats]

References

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A Appendix: conventions and fit quality

All yields are effective annual. Brazil's 252-business-day exponential quotes pass through unchanged; US and Mexican semiannual bond-equivalent yields compound to effective annual; Cetes convert from simple act/360 discount quotes; overnight act/360 policy quotes convert via $(1 + r/360)^{365} - 1$. Median fit RMSE by curve:

Curve	Days	Median RMSE bp
MX	1522	9.0
US	1638	3.5
BR	1494	3.4
DE	1529	published params

Table 6. Median Nelson–Siegel(–Svensson) fit RMSE by curve over the sample.

Appendix C of the companion desk note tabulates the forward-implied versus naive policy-path metric; the provenance table there maps every data number to its generating function.